

Eco Font Converter

Save up to 33% printer ink by converting documents to stroke-only text rendering — the same technique used by the Ryman Eco font.

Instead of filling each character solid, it draws only the outlines. The result looks identical on screen but uses significantly less ink when printed.

Requirements

- Python 3.10+
- pip

Install core dependencies:

```
pip install pypdf reportlab
```

Optional (only needed for those formats):

```
pip install python-docx # for .docx files pip install markdown # for .md files
```

Or let the script install them automatically with `-y`.

Usage

Single file

```
python eco_font_converter.py report.pdf python eco_font_converter.py notes.docx  
python eco_font_converter.py README.md
```

Output is saved next to the input as `<name>_eco.pdf`.

Custom output path

```
python eco_font_converter.py report.pdf --output report_print.pdf python  
eco_font_converter.py report.pdf -o report_print.pdf
```

Batch — multiple files or globs

```
python eco_font_converter.py *.pdf python eco_font_converter.py *.pdf *.docx *.md  
python eco_font_converter.py file1.pdf file2.docx
```

Batch — entire folder

```
python eco_font_converter.py --folder ./reports python eco_font_converter.py -f
./reports --output-dir ./eco_output
```

Processes all .pdf, .docx, .md, and .markdown files in the folder.

Auto-install dependencies

```
python eco_font_converter.py report.pdf --auto-install python eco_font_converter.py
report.pdf -y
```

Installs any missing packages via pip before converting.

Preview without converting

```
python eco_font_converter.py *.pdf --dry-run
```

Shows what files would be converted and where output would go, without touching anything.

Interactive mode

Run with no arguments and the script will prompt you to enter a file path:

```
python eco_font_converter.py
```

Useful when launching from a desktop shortcut or double-clicking the file.

All options

Flag	Short	Default	Description
--output	-o	<name>_eco.pdf	Output path (single file only)
--output-dir	-d	same folder as input	Output folder for batch mode
--folder	-f	—	Process all supported files in a folder
--stroke-width	-s	0.7	Thickness of character outlines (see below)
--auto-install	-y	off	Auto-install missing pip packages
--dry-run	—	off	Preview without converting

Stroke width guide

The stroke width controls how thick the character outlines are.

| Range | Effect |

|——|——|

| 0.4 - 0.6 | Very light — maximum ink saving, may look faint |

| 0.7 | Default — good balance of readability and savings |

| 0.8 - 1.0 | More visible — easier to read, less saving |

```
# Lighter (save more ink) python eco_font_converter.py report.pdf --stroke-width 0.5
# Heavier (easier to read) python eco_font_converter.py report.pdf --stroke-width 0.9
```

Supported formats

| Format | Notes |

|——|——|

| .pdf | Injects stroke-only rendering directly into the PDF stream |

| .docx | Extracts text and headings, renders to eco PDF |

| .md / .markdown | Parses Markdown including headings, bold, italic, code blocks, and lists |

How it works

PDF files contain content streams — sequences of drawing commands. Text is typically rendered with mode 0 (fill). This script:

Decompresses each content stream

Removes any existing text rendering mode commands

Wraps every BT...ET 0 Tr text block with (stroke-only mode) and sets a stroke width

Recompresses and writes the result

For DOCX and Markdown, the script first renders the content to a temporary PDF using ReportLab, then applies the same eco transformation.

Examples

```
# Convert one PDF, default settings python eco_font_converter.py quarterly_report.pdf
# Convert all PDFs in a folder, save to a separate output folder python
eco_font_converter.py --folder ./invoices --output-dir ./invoices_eco # Very light
strokes for maximum ink saving python eco_font_converter.py brochure.pdf -s 0.5 #
Convert a Word doc, auto-installing dependencies python eco_font_converter.py
```

```
proposal.docx -y # Preview what a glob would convert python eco_font_converter.py  
*.docx --dry-run
```